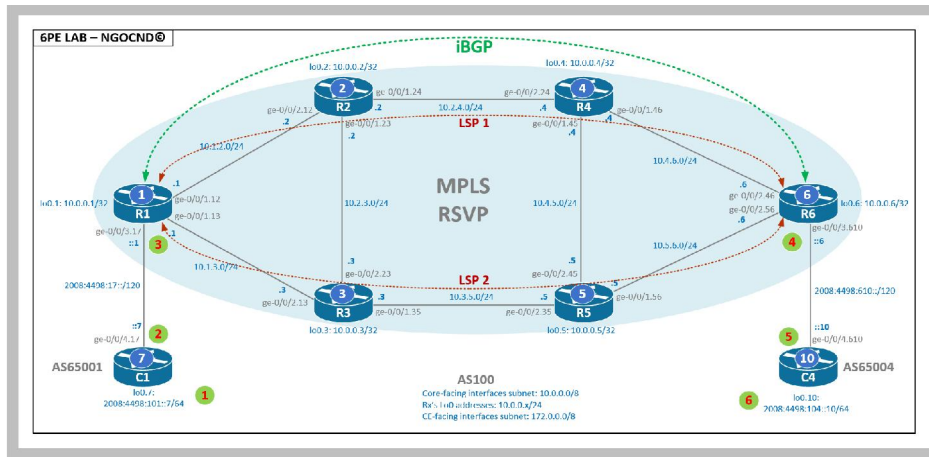


Contents

TOPOLOGY.....	2
SUMMARY RESULTS	3
A FEW WORDS	4
BASE CONFIGURATION INCLUDES:	5
TC1: IBGP FAMILY INET6 LABELED-UNICAST EXPLICIT-NULL	5
CONFIGURATION	5
VERIFICATION	6
PING TEST	13
TC2: IBGP FAMILY INET6 LABELED-UNICAST	16
CONFIGURATION	16
VERIFICATION	16
TC3: IBGP FAMILY INET6 LABELED-UNICAST + ENABLE FAMILY INET6 ON INTERFACES ON R2, R3 FACING R1 AND R4, R5 FACING R6	25
CONFIGURATION	25
VERIFICATION	25
PING TEST	25
TC4: IBGP FAMILY INET6 LABELED-UNICAST + MPLS LSP EXPLICIT-NULL	28
CONFIGURATION	28
VERIFICATION	28
PING TEST	28
TC5: IBGP FAMILY INET6 UNICAST	30
CONFIGURATION	30
VERIFICATION	30
PING TEST	37
TC6: IBGP FAMILY INET6 UNICAST + ENABLE FAMILY INET6 ON INTERFACES ON R2, R3 FACING R1 AND R4, R5 FACING R6	38
CONFIGURATION	38
VERIFICATION	39
PING TEST	39
TC7: IBGP FAMILY INET6 UNICAST + ENABLE INET6 ON R2,R3 FACING R1 AND R4,R5 FACING R6 + RSVP EXPLICIT - NULL.....	41
CONFIGURATION	41
VERIFICATION	41
PING TEST	41
TC8: IBGP FAMILY INET6 LABELED-UNICAST + C1, C4 WILL ADVERTISE DIRECTED IPV6 RANGE + ENABLE ADVERTISE-INACTIVE KNOB ON PE ROUTERS (R1, R6)	43
CONFIGURATION	43
VERIFICATION	44
PING TEST	53

Topology



This topology is built on one vMX running 14.1R1.10. Rx routers is logical-systems. C1, C4 are virtual routers on global of vMX.



03-6pe-initial.txt

Summary results

Test Case	Ping between CE1-CE2 (1) - (6)	Ping between CE1-CE2 (2) - (5)	Ping between PE1-PE2 (3) - (4)	Ping between PE1-CE2 (3) - (5)	Ping between PE1-CE2 (3) - (6)	Ping between PE2-CE1 (4) - (2)	Ping between PE2-CE1 (4) - (1)
TC1: iBGP family inet6 labeled-unicast explicit-null	OK	OK	OK	OK	OK	OK	OK
TC2: iBGP family inet6 labeled-unicast	OK	Failed	Failed	Failed	Failed	Failed	Failed
TC3: iBGP family inet6 labeled-unicast + enable inet6 on R2,R3 facing R1 and R4,R5 facing R6	OK	OK	OK	OK	OK	OK	OK
TC4: iBGP family inet6 labeled-unicast + RSVP explicit-null	OK	Failed	Failed	Failed	Failed	Failed	Failed
TC5: iBGP family inet6 unicast	Failed	Failed	Failed	Failed	Failed	Failed	Failed
TC6: iBGP family inet6 unicast + enable inet6 on R2,R3 facing R1 and R4,R5 facing R6	OK	OK	OK	OK	OK	OK	OK
TC7: iBGP family inet6 unicast + enable inet6 on R2,R3 facing R1 and R4,R5 facing R6 + RSVP explicit-null	Failed	Failed	Failed	Failed	Failed	Failed	Failed
TC8: iBGP family inet6 labeled-unicast + C1, C4 will advertise directed IPv6 range + enable advertise-inactive knob on PE routers (R1, R6)	OK	OK	OK	OK	OK	OK	OK

Commented [NND1]: Failed due to ipv6 packets is dropped at R2,R3 or R4,R5 after they executed pop the label and they observed these packets is ipv6 but out-going interface don't have inet6 enable, so these routers was dropped the packets.

Commented [NND2]: From result of TC2 and TC3, we can see that after PHP routers pop the label, they simply put the ipv6 packets out out-going interfaces towards R1/R6 even though PHP routers don't have these ipv6 routes in its inet6.0 table.

Commented [NND3]: Even these packets reach to R1 or R6 but they still failed due to once R1 or R6 pop the label 0, they observe its payload inside is ipv6, but label 0 just only using for ipv4 payload, so R1/R6 routers drop these packets. IMPORTANT NOTES: end-to-end traffic (1)-(6) don't using label 0 for transport, due to end-to-end traffic have its particular label so transport label can't use label 0 (label 0 can't use for stack label).

Commented [NND4]: All ipv6 packets go through core use only one transport label. Similar to TC2, failed due to ipv6 packets is dropped at R2,R3 or R4,R5 after these routers executed pop the label and they observed these packets is ipv6 but out-going interface don't have inet6 enable, so these routers was dropped the packets.

Commented [NND5]: Similar to TC3, after PHP (R2,3,4,5) routers pop the label, they simply put the ipv6 packets out out-going interfaces towards R1/R6 even though PHP routers don't have these ipv6 routes in its inet6.0 table.

Commented [NND6]: All packets using MPLS LSP must pop at egress (R1/R6) router. So similar to TC4, once R1 or R6 pop the label 0, they observe its payload inside is ipv6, but label 0 just only using for ipv4 payload, so R1/R6 routers drop these packets.

Commented [NND7]: Once we let C devices (C1, C4) advertise directed IPv6 range rather than PE routers (R1, R6), so when R1/R6 receive these routes, they generate label for these routes like routes from C devices, normal label, not special label "3", so packets reach to R1/R6 routers still have bottom label, so they can reachable from directed connections.

A FEW WORDS

From the results of these test cases above, for IPv6 reachability purpose via ISP backbone purely running ipv4, we can use `inet6 labeled-unicast` or `inet6 unicast` family on iBGP session. There are a few considerations that we have to know while you make a decision as below:

1/. If you choose `inet6 labeled-unicast`, **all packets go through ISP backbone will use label stack except direction connect**, e.g: ping between (2) and (5). Actually we don't have to enable `explicit-null` on family `inet6 labeled-unicast`. If we want to be reachable between directed connections, we can be done by doing one of these below options:

1.1/. Enable `explicit-null` on family `inet6 labeled-unicast` as with TC1

1.2/. On P routers, enable family `inet6` on interfaces that towards to PE routers as with TC3

1.3/. PE's routers don't advertise directed IPv6 range PE-CE, CE devices will do it. In this case, once PE routers receive this routes from CE routers, it can't advertise to remote PE routers due to this IPv6 range is inactive with BGP due to it's direct, so PE routers must enable `advertise-inactive` to permit PE routers proceed advertising this IPv6 range to remote PE router as show in TC8

2/. If you choose `inet6 unicast`, you must ensure LDP/RSVP do NOT enable `explicit-null` and you have to enable family `inet6` on interfaces that towards to PE routers on P routers. All packets go through ISP backbone will use only one label for transport and must be popped by PHP routers.

Base configuration includes:

1. Interface/IP configure as topology
2. Native IPv6 eBGP between C1-R1 and C4-R6
 - o C1: advertise a summary route: 2008:4498:101::/48
 - o C4: advertise a summary route: 2008:4498:104::/48
3. Infrastructure:
 - o OSPF area0 between internal Rx routers
 - o MPLS-RSVP running between Rx routers as shown in the diagram
 - o 2 MPLS LSPs running between R1 and R6
 - LSP 1 running top path
 - LSP 2 running bottom path
4. We'll also advertise ipv6 directed range between R1-C1 and R6-C4 for each other but on the base configuration we don't define this policy yet, we'll define on TC1 and will use for overall these tests, except TC8.

TC1: iBGP family inet6 labeled-unicast explicit-null

Configuration

1. Enable MPLS ipv6-tunnelling on R1 and R6:

```
lab@vMX3# sh|compare
[edit logical-systems r1 protocols mpls]
+   ipv6-tunneling;
[edit logical-systems r6 protocols mpls]
+   ipv6-tunneling;
```

2. Enable family inet6 on all core-facing interfaces on R1 and R6

```
lab@vMX3# show | compare
[edit logical-systems r1 interfaces ge-0/0/1 unit 12]
+   family inet6;
[edit logical-systems r1 interfaces ge-0/0/1 unit 13]
+   family inet6;
[edit logical-systems r6 interfaces ge-0/0/2 unit 46]
+   family inet6;
[edit logical-systems r6 interfaces ge-0/0/2 unit 56]
+   family inet6;
```

3. Configure iBGP between R1 and R6 support family inet6 label unicast explicit-null

```
lab@vMX3# sh|compare
[edit logical-systems r1 protocols bgp]
  group ebgp-c1-v6 { ... }
+   group ibgp {
+     type internal;
+     local-address 10.0.0.1;
+     family inet6 {
+       labeled-unicast {
+         explicit-null;
+       }
+     }
+     neighbor 10.0.0.6;
+   }
[edit logical-systems r6 protocols bgp]
  group ebgp-c4-v6 { ... }
+   group ibgp {
+     type internal;
+     local-address 10.0.0.6;
+     family inet6 {
+       labeled-unicast {
+         explicit-null;
+       }
+     }
+   }
```

```
+ neighbor 10.0.0.1;
+ }
```

4. Redistribute range directed ipv6 R1-C1 and R6-C4 for each other

```
lab@vMX3# sh|compare
[edit logical-systems r1 protocols bgp group ibgp]
+ export bgp-exp;
[edit logical-systems r1]
+ policy-options {
+   policy-statement bgp-exp {
+     term 1 {
+       from {
+         protocol direct;
+         rib inet6.0;
+         route-filter 2008:4498::/32 longer;
+       }
+       then accept;
+     }
+   }
+ }
[edit logical-systems r6 protocols bgp group ibgp]
+ export bgp-exp;
[edit logical-systems r6 policy-options]
+ policy-statement bgp-exp {
+   term 1 {
+     from {
+       protocol direct;
+       rib inet6.0;
+       route-filter 2008:4498::/32 longer;
+     }
+     then accept;
+   }
+ }
```

Verification

On R1 router:

```
lab@vMX3# run show route logical-system r1 table inet.3

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

10.0.0.6/32          *[RSVP/7/1] 15:10:07, metric 3
                    > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-1
                    > to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-2

[edit]
lab@vMX3# run show route logical-system r1 table inet6.3

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

::ffff:10.0.0.6/128* [RSVP/7/1] 00:21:25, metric 3
                    > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-1
                    > to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-2

[edit]
lab@vMX3# run show bgp summary logical-system r1
Groups: 2 Peers: 2 Down peers: 0
Table      Tot Paths  Act Paths Suppressed    History Damp State    Pending
inet.0
          0         0         0         0         0         0
inet6.0
          3         3         0         0         0         0
```

Peer	AS	InPkt	OutPkt	OutQ	Flaps	Last Up/Dwn
State #Active/Received/Accepted/Damped...						
10.0.0.6	100	16	17	0	0	5:29 Establ
inet6.0: 2/2/2/0						
2008:4498:17::7	65001	791	790	0	2	5:55:31 Establ
inet6.0: 1/1/1/0						

```

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1 extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 2
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] I

* 2008:4498:101::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 2
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] 65001 I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.6 logical-system r1 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

mpls.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  Accepted
  Route Label: 2
  Nexthop: ::ffff:10.0.0.6
  Localpref: 100
  AS path: 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
  Accepted
  Route Label: 2
  Nexthop: ::ffff:10.0.0.6
  Localpref: 100
  AS path: I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 logical-system r1 protocol bgp

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:101::/64 *[BGP/170] 06:00:27, localpref 100
  AS path: 65001 I, validation-state: unverified
  > to 2008:4498:17::7 via ge-0/0/3.17
2008:4498:104::/64 *[BGP/170] 00:10:25, localpref 100, from 10.0.0.6
  AS path: 65004 I, validation-state: unverified
  > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-1
  > to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-2
2008:4498:610::/120*[BGP/170] 00:07:23, localpref 100, from 10.0.0.6
  AS path: I, validation-state: unverified
  > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-1

```

to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-2

```
[edit]
lab@vMX3# run show route table inet6.0 logical-system r1 protocol bgp extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
2008:4498:101::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:101::/64 -> {2008:4498:17::7}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a7d0 (adv_entry)
  Advertised metrics:
    Flags: Nexthop Change
    Nexthop: Self
    Localpref: 100
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:101:: from 2008:4498:17::7 Vector len 4. Val: 1
  *BGP Preference: 170/-101
    Next hop type: Router, Next hop index: 1266
    Address: 0x956010c
    Next-hop reference count: 3
    Source: 2008:4498:17::7
    Next hop: 2008:4498:17::7 via ge-0/0/3.17, selected
    Session Id: 0x380003
    State: <Active Ext>
    Local AS: 100 Peer AS: 65001
    Age: 6:00:36
    Validation State: unverified
    Task: BGP_65001.2008:4498:17::7+63777
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65001 I
    Accepted
    Localpref: 100
    Router ID: 101.0.0.7

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {indirect(1048578)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a8cc (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65004 I
    Communities:
Path 2008:4498:104:: from 10.0.0.6 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9560844
    Next-hop reference count: 6
    Source: 10.0.0.6
    Next hop type: Router, Next hop index: 1048577
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
    Label-switched-path LSP-R1-to-R6-1
    Label operation: Push 2, Push 300080(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 2: None; Label 300080: None;
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Label-switched-path LSP-R1-to-R6-2
    Label operation: Push 2, Push 299984(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 2: None; Label 299984: None;
    Session Id: 0x380005
    Protocol next hop: ::ffff:10.0.0.6
    Label operation: Push 2
    Label TTL action: prop-ttl
    Load balance label: Label 2: None;
    Indirect next hop: 0x976d918 1048578 INH Session ID: 0x38000a
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
```

```

Age: 10:34      Metric2: 3
Validation State: unverified
Task: BGP 100.10.0.0.6+65317
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: 65004 I
Accepted
Route Label: 2
Localpref: 100
Router ID: 10.0.0.6
Indirect next hops: 1
    Protocol next hop: ::ffff:10.0.0.6 Metric: 3
    Label operation: Push 2
    Label TTL action: prop-ttl
    Load balance label: Label 2: None;
    Indirect next hop: 0x976d918 1048578 INH Session ID: 0x38000a
    Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
        Session Id: 0x380004
        Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
        Session Id: 0x380005
        ::ffff:10.0.0.6/128 Originating RIB: inet6.3
        Metric: 3                      Node path count: 1
        Forwarding nexthops: 2
        Nexthop: 10.1.2.2 via ge-0/0/1.12

2008:4498:610::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:610::/120 -> {indirect(1048578)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a728 (adv_entry)
    Advertised metrics:
        Nexthop: Self
        AS path: [100] I
        Communities:
Path 2008:4498:610:: from 10.0.0.6 Vector len 4. Val: 0
    *BGP      Preference: 170/-101
        Next hop type: Indirect
        Address: 0x9560844
        Next-hop reference count: 6
        Source: 10.0.0.6
        Next hop type: Router, Next hop index: 1048577
        Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
        Label-switched-path LSP-R1-to-R6-1
        Label operation: Push 2, Push 300080(top)
        Label TTL action: prop-ttl, prop-ttl(top)
        Load balance label: Label 2: None; Label 300080: None;
        Session Id: 0x380004
        Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
        Label-switched-path LSP-R1-to-R6-2
        Label operation: Push 2, Push 299984(top)
        Label TTL action: prop-ttl, prop-ttl(top)
        Load balance label: Label 2: None; Label 299984: None;
        Session Id: 0x380005
        Protocol next hop: ::ffff:10.0.0.6
        Label operation: Push 2
        Label TTL action: prop-ttl
        Load balance label: Label 2: None;
        Indirect next hop: 0x976d918 1048578 INH Session ID: 0x38000a
        State: <Active Int Ext>
        Local AS: 100 Peer AS: 100
        Age: 7:32      Metric2: 3
        Validation State: unverified
        Task: BGP 100.10.0.0.6+65317
        Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
        AS path: I
        Accepted
        Route Label: 2
        Localpref: 100
        Router ID: 10.0.0.6

```

```

Indirect next hops: 1
  Protocol next hop: ::ffff:10.0.0.6 Metric: 3
  Label operation: Push 2
  Label TTL action: prop-ttl
  Load balance label: Label 2: None;
  Indirect next hop: 0x976d918 1048578 INH Session ID: 0x38000a
  Indirect path forwarding next hops: 2
    Next hop type: Router
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Session Id: 0x380005
  ::ffff:10.0.0.6/128 Originating RIB: inet6.3
  Metric: 3                      Node path count: 1
  Forwarding nexthops: 2
    Nexthop: 10.1.2.2 via ge-0/0/1.12

```

On R6 router:

```

lab@vMX3# run show route logical-system r6 table inet.3

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

10.0.0.1/32          *[RSVP/7/1] 15:10:48, metric 3
                    to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
                    > to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2

[edit]
lab@vMX3# run show route logical-system r6 table inet6.3

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

::ffff:10.0.0.1/128*[RSVP/7/1] 00:22:28, metric 3
                    to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
                    > to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2

[edit]
lab@vMX3# run show bgp summary logical-system r6
Groups: 2 Peers: 2 Down peers: 0
Table      Tot Paths  Act Paths Suppressed    History Damp State    Pending
inet6.0
          3          3          0          0          0          0
inet.0
          0          0          0          0          0          0
Peer      AS      InPkt    OutPkt    OutQ    Flaps Last Up/Dwn
State|#Active/Received/Accepted/Damped...
10.0.0.1  100      41       41        0        0      17:06 Establ
  inet6.0: 2/2/2/0
2008:4498:610::10 65004    816      821        0        2      6:07:05 Establ
  inet6.0: 1/1/1/0

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6 extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 2
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 2
    Nexthop: Self

```

```

Flags: Nexthop Change
Localpref: 100
AS path: [100] I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.1 logical-system r6 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)
inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
mpls.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)
inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  Accepted
  Route Label: 2
  Nexthop: ::ffff:10.0.0.1
  Localpref: 100
  AS path: I
* 2008:4498:101::/64 (1 entry, 1 announced)
  Accepted
  Route Label: 2
  Nexthop: ::ffff:10.0.0.1
  Localpref: 100
  AS path: 65001 I
inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 logical-system r6 protocol bgp

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:17::/120 * [BGP/170] 00:17:40, localpref 100, from 10.0.0.1
  AS path: I, validation-state: unverified
  > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
  to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2
2008:4498:101::/64 * [BGP/170] 00:20:42, localpref 100, from 10.0.0.1
  AS path: 65001 I, validation-state: unverified
  > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
  to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2
2008:4498:104::/64 * [BGP/170] 06:10:41, localpref 100
  AS path: 65004 I, validation-state: unverified
  > to 2008:4498:610::10 via ge-0/0/3.610

[edit]
lab@vMX3# run show route table inet6.0 logical-system r6 protocol bgp extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
2008:4498:17::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:17::/120 -> {indirect(1048576)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957a2ac (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] I
  Communities:
Path 2008:4498:17:: from 10.0.0.1 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x95636dc
    Next-hop reference count: 6
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048575
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
    Label-switched-path LSP-R6-to-R1-1

```

```

Label operation: Push 2, Push 300016(top)
Label TTL action: prop-ttl, prop-ttl(top)
Load balance label: Label 2: None; Label 300016: None;
Session Id: 0x300005
Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
Label-switched-path LSP-R6-to-R1-2
Label operation: Push 2, Push 299952(top)
Label TTL action: prop-ttl, prop-ttl(top)
Load balance label: Label 2: None; Label 299952: None;
Session Id: 0x300004
Protocol next hop: ::ffff:10.0.0.1
Label operation: Push 2
Label TTL action: prop-ttl
Load balance label: Label 2: None;
Indirect next hop: 0x9759918 1048576 INH Session ID: 0x30000d
State: <Active Int Ext>
Local AS: 100 Peer AS: 100
Age: 17:44 Metric2: 3
Validation State: unverified
Task: BGP_100.10.0.0.1+179
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: I
Accepted
Route Label: 2
Localpref: 100
Router ID: 10.0.0.1
Indirect next hops: 1
    Protocol next hop: ::ffff:10.0.0.1 Metric: 3
    Label operation: Push 2
    Label TTL action: prop-ttl
    Load balance label: Label 2: None;
    Indirect next hop: 0x9759918 1048576 INH Session ID: 0x30000d
    Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%
        Session Id: 0x300005
        Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
        Session Id: 0x300004
    ::ffff:10.0.0.1/128 Originating RIB: inet6.3
    Metric: 3 Node path count: 1
    Forwarding nexthops: 2
    Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:101::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:101::/64 -> {indirect(1048576)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957abc0 (adv_entry)
    Advertised metrics:
        Nexthop: Self
        AS path: [100] 65001 I
        Communities:
Path 2008:4498:101:: from 10.0.0.1 Vector len 4. Val: 0
    *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x95636dc
    Next-hop reference count: 6
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048575
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
    Label-switched-path LSP-R6-to-R1-1
    Label operation: Push 2, Push 300016(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 2: None; Label 300016: None;
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Label-switched-path LSP-R6-to-R1-2
    Label operation: Push 2, Push 299952(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 2: None; Label 299952: None;

```

```

Session Id: 0x300004
Protocol next hop: ::ffff:10.0.0.1
Label operation: Push 2
Label TTL action: prop-ttl
Load balance label: Label 2: None;
Indirect next hop: 0x9759918 1048576 INH Session ID: 0x30000d
State: <Active Int Ext>
Local AS: 100 Peer AS: 100
Age: 20:46 Metric2: 3
Validation State: unverified
Task: BGP_100.10.0.0.1+179
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: 65001 I
Accepted
Route Label: 2
Localpref: 100
Router ID: 10.0.0.1
Indirect next hops: 1
    Protocol next hop: ::ffff:10.0.0.1 Metric: 3
    Label operation: Push 2
    Label TTL action: prop-ttl
    Load balance label: Label 2: None;
    Indirect next hop: 0x9759918 1048576 INH Session ID: 0x30000d
    Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%
        Session Id: 0x300005
        Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
        Session Id: 0x300004
    ::ffff:10.0.0.1/128 Originating RIB: inet6.3
        Metric: 3 Node path count: 1
        Forwarding nexthops: 2
            Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {2008:4498:610::10}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a744 (adv_entry)
    Advertised metrics:
        Flags: Nexthop Change
        Nexthop: Self
        Localpref: 100
        AS path: [100] 65004 I
        Communities:
Path 2008:4498:104:: from 2008:4498:610::10 Vector len 4. Val: 1
    *BGP Preference: 170/-101
        Next hop type: Router, Next hop index: 1267
        Address: 0x9560550
        Next-hop reference count: 3
        Source: 2008:4498:610::10
        Next hop: 2008:4498:610::10 via ge-0/0/3.610, selected
        Session Id: 0x300003
        State: <Active Ext>
        Local AS: 100 Peer AS: 65004
        Age: 6:10:45
        Validation State: unverified
        Task: BGP_65004.2008:4498:610::10+57753
        Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
        AS path: 65004 I
        Accepted
        Localpref: 100
        Router ID: 104.0.0.10

```

[edit]

Ping test

```
lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)
```

```
[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=60 time=8.653 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.653/8.653/8.653/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=60 time=10.739 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.739/10.739/10.739/0.000 ms

[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=60 time=10.237 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.237/10.237/10.237/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=60 time=9.861 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.861/9.861/9.861/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=64 time=7.116 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.116/7.116/7.116/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=64 time=8.462 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.462/8.462/8.462/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
```

```

lab@vMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=63 time=10.094 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.094/10.094/10.094/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=5.341 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 5.341/5.341/5.341/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=63 time=7.273 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.273/7.273/7.273/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=9.038 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.038/9.038/9.038/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=63 time=8.513 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.513/8.513/8.513/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=13.909 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 13.909/13.909/13.909/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1

```

```

PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=63 time=11.079 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 11.079/11.079/11.079/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=9.913 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.913/9.913/9.913/0.000 ms

```

TC2: iBGP family inet6 labeled-unicast

Configuration

Continue with the configuration on TC1 and remove explicit-null on family inet6 labeled-unicast on R1 and R6

```

lab@vMX3# sh|compare
[edit logical-systems r1 protocols bgp group ibgp family inet6 labeled-unicast]
- explicit-null;
[edit logical-systems r6 protocols bgp group ibgp family inet6 labeled-unicast]
- explicit-null;

```

Verification

On R1 router

```

lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
  Prefix            Nexthop          MED      Lclpref    AS path
* 2008:4498:17::/120 Self              100       100        I
* 2008:4498:101::/64 Self              100       100        65001 I

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1
extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 3
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] I

* 2008:4498:101::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 300064
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] 65001 I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.6 logical-system r1 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

```

```

mpls.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  Accepted
  Route Label: 299904
  Nexthop: ::ffff:10.0.0.6
  Localpref: 100
  AS path: 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
  Accepted
  Route Label: 3
  Nexthop: ::ffff:10.0.0.6
  Localpref: 100
  AS path: I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system rl

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:101::/64 *[BGP/170] 06:22:22, localpref 100
    AS path: 65001 I, validation-state: unverified
    > to 2008:4498:17::7 via ge-0/0/3.17
2008:4498:104::/64 *[BGP/170] 00:03:37, localpref 100, from 10.0.0.6
    AS path: 65004 I, validation-state: unverified
    > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-
1
    to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-
2
2008:4498:610::/120*[BGP/170] 00:03:37, localpref 100, from 10.0.0.6
    AS path: I, validation-state: unverified
    > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-
1
    to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-
2

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system rl extensive |
no-more

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
2008:4498:101::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:101::/64 -> {2008:4498:17::7}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a7d0 (adv_entry)
  Advertised metrics:
    Flags: Nexthop Change
    Nexthop: Self
    Localpref: 100
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:101:: from 2008:4498:17::7 Vector len 4. Val: 1
  *BGP Preference: 170/-101
    Next hop type: Router, Next hop index: 1266
    Address: 0x956010c
    Next-hop reference count: 4
    Source: 2008:4498:17::7
    Next hop: 2008:4498:17::7 via ge-0/0/3.17, selected
    Session Id: 0x380003
    State: <Active Ext>
    Local AS: 100 Peer AS: 65001
    Age: 6:22:26
    Validation State: unverified

```

```

Task: BGP_65001.2008:4498:17::7+63777
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: 65001 I
Accepted
Localpref: 100
Router ID: 101.0.0.7

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {indirect(1048582)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a8cc (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65004 I
    Communities:
Path 2008:4498:104:: from 10.0.0.6 Vector len 4. Val: 0
  *BGP
    Preference: 170/-101
    Next hop type: Indirect
    Address: 0x95633e8
    Next-hop reference count: 3
    Source: 10.0.0.6
    Next hop type: Router, Next hop index: 1048581
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
    Label-switched-path LSP-R1-to-R6-1
    Label operation: Push 299904, Push 300080(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 299904: None; Label 300080: None;
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Label-switched-path LSP-R1-to-R6-2
    Label operation: Push 299904, Push 299984(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 299904: None; Label 299984: None;
    Session Id: 0x380005
    Protocol next hop: ::ffff:10.0.0.6
    Label operation: Push 299904
    Label TTL action: prop-ttl
    Load balance label: Label 299904: None;
    Indirect next hop: 0x976da3c 1048582 INH Session ID: 0x38000a
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 3:41 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.6+65317
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65004 I
    Accepted
    Route Label: 299904
    Localpref: 100
    Router ID: 10.0.0.6
    Indirect next hops: 1
      Protocol next hop: ::ffff:10.0.0.6 Metric: 3
      Label operation: Push 299904
      Label TTL action: prop-ttl
      Load balance label: Label 299904: None;
      Indirect next hop: 0x976da3c 1048582 INH Session ID:
0x38000a
      Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
        Session Id: 0x380004
        Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
        Session Id: 0x380005
        ::ffff:10.0.0.6/128 Originating RIB: inet6.3
        Metric: 3 Node path count: 1
        Forwarding nexthops: 2
        Nexthop: 10.1.2.2 via ge-0/0/1.12

2008:4498:610::/120 (1 entry, 1 announced)

```

```

TSI:
KRT in-kernel 2008:4498:610::/120 -> {indirect(1048585)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a728 (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] I
    Communities:
Path 2008:4498:610:: from 10.0.0.6 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9560748
    Next-hop reference count: 3
    Source: 10.0.0.6
    Next hop type: Router, Next hop index: 1048583
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
    Label-switched-path LSP-R1-to-R6-1
    Label operation: Push 300080
    Label TTL action: prop-ttl
    Load balance label: Label 300080: None;
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Label-switched-path LSP-R1-to-R6-2
    Label operation: Push 299984
    Label TTL action: prop-ttl
    Load balance label: Label 299984: None;
    Session Id: 0x380005
    Protocol next hop: ::ffff:10.0.0.6
    Indirect next hop: 0x976db60 1048585 INH Session ID: 0x38000a
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 3:41 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.6+65317
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: I
    Accepted
    Route Label: 3
    Localpref: 100
    Router ID: 10.0.0.6
    Indirect next hops: 1
      Protocol next hop: ::ffff:10.0.0.6 Metric: 3
      Indirect next hop: 0x976db60 1048585 INH Session ID:
0x38000a
      Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
        Session Id: 0x380004
        Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
        Session Id: 0x380005
        ::ffff:10.0.0.6/128 Originating RIB: inet6.3
        Metric: 3 Node path count: 1
        Forwarding nexthops: 2
        Nexthop: 10.1.2.2 via ge-0/0/1.12
lab@vMX3# run show route table mpls.0 logical-system r1

mpls.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

0          *[MPLS/0] 15:33:30, metric 1
           Receive
1          *[MPLS/0] 15:33:30, metric 1
           Receive
2          *[MPLS/0] 15:33:30, metric 1
           Receive
13         *[MPLS/0] 15:33:30, metric 1
           Receive
300064     *[VPN/170] 00:13:32
           > to 2008:4498:17::7 via ge-0/0/3.17, Pop

```

[edit]

On R6 router

```
lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
  Prefix          Nexthop          MED      Lclpref    AS path
* 2008:4498:104::/64  Self              100        100      65004 I
* 2008:4498:610::/120 Self              100        100        I

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6
extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 299904
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 3
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.1 logical-system r6 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

mpls.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  Accepted
    Route Label: 3
    Nexthop: ::ffff:10.0.0.1
    Localpref: 100
    AS path: I

* 2008:4498:101::/64 (1 entry, 1 announced)
  Accepted
    Route Label: 300064
    Nexthop: ::ffff:10.0.0.1
    Localpref: 100
    AS path: 65001 I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system r6

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:17::/120 *[BGP/170] 00:08:20, localpref 100, from 10.0.0.1
                   AS path: I, validation-state: unverified
                   > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-
1
                   to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-
2
```

```

2008:4498:101::/64 *[BGP/170] 00:08:20, localpref 100, from 10.0.0.1
    AS path: 65001 I, validation-state: unverified
    > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-
1
    to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-
2
2008:4498:104::/64 *[BGP/170] 06:27:02, localpref 100
    AS path: 65004 I, validation-state: unverified
    > to 2008:4498:610::10 via ge-0/0/3.610

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system r6 extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
2008:4498:17::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:17::/120 -> {indirect(1048586)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957a2ac (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] I
    Communities:
Path 2008:4498:17:: from 10.0.0.1 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9562524
    Next-hop reference count: 3
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048584
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
    Label-switched-path LSP-R6-to-R1-1
    Label operation: Push 300016
    Label TTL action: prop-ttl
    Load balance label: Label 300016: None;
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Label-switched-path LSP-R6-to-R1-2
    Label operation: Push 299952
    Label TTL action: prop-ttl
    Load balance label: Label 299952: None;
    Session Id: 0x300004
    Protocol next hop: ::ffff:10.0.0.1
    Indirect next hop: 0x9759b60 1048586 INH Session ID: 0x30000d
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 8:23 Metric2: 3
    Validation State: unverified
    Task: BGP 100.10.0.0.1+179
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: I
    Accepted
    Route Label: 3
    Localpref: 100
    Router ID: 10.0.0.1
    Indirect next hops: 1
      Protocol next hop: ::ffff:10.0.0.1 Metric: 3
      Indirect next hop: 0x9759b60 1048586 INH Session ID:
0x30000d
      Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1
balance 33%
        Session Id: 0x300005
        Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1
balance 67%
        Session Id: 0x300004
        ::ffff:10.0.0.1/128 Originating RIB: inet6.3
        Metric: 3 Node path count: 1
        Forwarding nexthops: 2

```

```

Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:101::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:101::/64 -> {indirect(1048580)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957abc0 (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:101:: from 10.0.0.1 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9560a90
    Next-hop reference count: 3
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048579
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
    Label-switched-path LSP-R6-to-R1-1
    Label operation: Push 300064, Push 300016(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 300064: None; Label 300016: None;
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Label-switched-path LSP-R6-to-R1-2
    Label operation: Push 300064, Push 299952(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 300064: None; Label 299952: None;
    Session Id: 0x300004
    Protocol next hop: ::ffff:10.0.0.1
    Label operation: Push 300064
    Label TTL action: prop-ttl
    Load balance label: Label 300064: None;
    Indirect next hop: 0x9759a3c 1048580 INH Session ID: 0x30000d
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 8:23 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.1+179
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65001 I
    Accepted
    Route Label: 300064
    Localpref: 100
    Router ID: 10.0.0.1
    Indirect next hops: 1
      Protocol next hop: ::ffff:10.0.0.1 Metric: 3
      Label operation: Push 300064
      Label TTL action: prop-ttl
      Load balance label: Label 300064: None;
      Indirect next hop: 0x9759a3c 1048580 INH Session ID:
0x30000d
      Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1
balance 33%
        Session Id: 0x300005
        Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1
balance 67%
        Session Id: 0x300004
        ::ffff:10.0.0.1/128 Originating RIB: inet6.3
        Metric: 3 Node path count: 1
        Forwarding nexthops: 2
        Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {2008:4498:610::10}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a744 (adv entry)

```

```

Advertised metrics:
  Flags: Nexthop Change
  Nexthop: Self
  Localpref: 100
  AS path: [100] 65004 I
  Communities:
Path 2008:4498:104:: from 2008:4498:610::10 Vector len 4. Val: 1
  *BGP      Preference: 170/-101
            Next hop type: Router, Next hop index: 1267
            Address: 0x9560550
            Next-hop reference count: 4
            Source: 2008:4498:610::10
            Next hop: 2008:4498:610::10 via ge-0/0/3.610, selected
            Session Id: 0x300003
            State: <Active Ext>
            Local AS: 100 Peer AS: 65004
            Age: 6:27:05
            Validation State: unverified
            Task: BGP_65004.2008:4498:610::10+57753
            Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
            AS path: 65004 I
            Accepted
            Localpref: 100
            Router ID: 104.0.0.10
lab@vMX3# run show route table mpls.0 logical-system r6

mpls.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

0          *[MPLS/0] 15:33:18, metric 1
            Receive
1          *[MPLS/0] 15:33:18, metric 1
            Receive
2          *[MPLS/0] 15:33:18, metric 1
            Receive
13         *[MPLS/0] 15:33:18, metric 1
            Receive
299904     *[VPN/170] 00:13:20
            > to 2008:4498:610::10 via ge-0/0/3.610, Pop

[edit]

```

Ping test

```

lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=60 time=10.191 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.191/10.191/10.191/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=60 time=7.703 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.703/7.703/7.703/0.000 ms

[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

```

```

[edit]
lab@VMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@VMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
lab@VMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@VMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---

```

```

1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

```

TC3: iBGP family inet6 labeled-unicast + enable family inet6 on interfaces on R2, R3 facing R1 and R4, R5 facing R6

Configuration

Continue with the configuration on TC2 and enable family inet6 on interfaces on R2, R3 facing R1 and R4, R5 facing R6

```

lab@vMX3# show | compare
[edit logical-systems r2 interfaces ge-0/0/2 unit 12]
+   family inet6;
[edit logical-systems r3 interfaces ge-0/0/2 unit 13]
+   family inet6;
[edit logical-systems r4 interfaces ge-0/0/1 unit 46]
+   family inet6;
[edit logical-systems r5 interfaces ge-0/0/1 unit 56]
+   family inet6;

```

Verification

The same on TC2 so we'll bypass this step for briefly

Ping test

```

lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

```

```
[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=60 time=10.815 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.815/10.815/10.815/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=60 time=9.784 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.784/9.784/9.784/0.000 ms

[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=60 time=13.492 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 13.492/13.492/13.492/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=60 time=9.968 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.968/9.968/9.968/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=62 time=10.059 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.059/10.059/10.059/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=62 time=10.032 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.032/10.032/10.032/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
```

```

lab@vMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=61 time=9.989 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.989/9.989/9.989/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=10.029 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.029/10.029/10.029/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=61 time=8.832 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.832/8.832/8.832/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=10.300 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.300/10.300/10.300/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=61 time=9.493 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.493/9.493/9.493/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=7.119 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.119/7.119/7.119/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1

```

```

PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=61 time=6.983 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 6.983/6.983/6.983/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=11.571 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 11.571/11.571/11.571/0.000 ms

```

TC4: iBGP family inet6 labeled-unicast + MPLS LSP explicit-null

Configuration

Continue with the configuration on TC3 but removing family inet6 on interfaces on R2, R3 facing R1 and R4, R5 facing R6 and configure MPLS LSP explicit-null on R1 and R6.

```

lab@vMX3# show | compare
[edit logical-systems r1 protocols mpls]
+   explicit-null;
[edit logical-systems r2 interfaces ge-0/0/2 unit 12]
-   family inet6;
[edit logical-systems r3 interfaces ge-0/0/2 unit 13]
-   family inet6;
[edit logical-systems r4 interfaces ge-0/0/1 unit 46]
-   family inet6;
[edit logical-systems r5 interfaces ge-0/0/1 unit 56]
-   family inet6;
[edit logical-systems r6 protocols mpls]
+   explicit-null;

```

Verification

The same on TC2 so we'll bypass this step for briefly

Ping test

```

lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=60 time=11.631 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 11.631/11.631/11.631/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=60 time=10.785 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.785/10.785/10.785/0.000 ms

```

```
[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1
```

```

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

```

TC5: iBGP family inet6 unicast

Configuration

Continue with the configuration on TC4 but removing MPLS LSP explicit-null and change ibgp family from inet6 labeled-unicast to inet6 unicast on R1 and R6.

```

lab@vMX3# show | compare
[edit logical-systems r1 protocols mpls]
-   explicit-null;
[edit logical-systems r1 protocols bgp group ibgp family inet6]
+   unicast;
-   labeled-unicast;
[edit logical-systems r6 protocols mpls]
-   explicit-null;
[edit logical-systems r6 protocols bgp group ibgp family inet6]
+   unicast;
-   labeled-unicast;

```

Verification

On R1 router

```
lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1
```

```

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
  Prefix      Nexthop      MED      Lclpref      AS path
* 2008:4498:17::/120      Self      100      I
* 2008:4498:101::/64      2008:4498:17::7      100      65001 I

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1 extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Nexthop: Self
    Localpref: 100
    AS path: [100] I

* 2008:4498:101::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Nexthop: 2008:4498:17::7
    Localpref: 100
    AS path: [100] 65001 I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.6 logical-system r1 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

mpls.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  Accepted
    Nexthop: 2008:4498:610::10
    Localpref: 100
    AS path: 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
  Accepted
    Nexthop: ::ffff:10.0.0.6
    Localpref: 100
    AS path: I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 logical-system r1 protocol bgp

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:101::/64 *[BGP/170] 08:11:37, localpref 100
  AS path: 65001 I, validation-state: unverified
    > to 2008:4498:17::7 via ge-0/0/3.17
2008:4498:104::/64 *[BGP/170] 00:14:42, localpref 100, from 10.0.0.6
  AS path: 65004 I, validation-state: unverified
    > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-1
    > to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-2
2008:4498:610::/120*[BGP/170] 00:14:42, localpref 100, from 10.0.0.6
  AS path: I, validation-state: unverified
    > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-1
    > to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-2

[edit]
lab@vMX3# run show route table inet6.0 logical-system r1 protocol bgp extensive | no-more

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
2008:4498:101::/64 (1 entry, 1 announced)
TSI:

```

```

KRT in-kernel 2008:4498:101::/64 -> {2008:4498:17::7}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a4dc (adv_entry)
  Advertised metrics:
    Nexthop: 2008:4498:17::7
    Localpref: 100
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:101:: from 2008:4498:17::7 Vector len 4. Val: 1
  *BGP
    Preference: 170/-101
    Next hop type: Router, Next hop index: 1266
    Address: 0x956010c
    Next-hop reference count: 3
    Source: 2008:4498:17::7
    Next hop: 2008:4498:17::7 via ge-0/0/3.17, selected
    Session Id: 0x380003
    State: <Active Ext>
    Local AS: 100 Peer AS: 65001
    Age: 8:11:42
    Validation State: unverified
    Task: BGP_65001.2008:4498:17::7+63777
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65001 I
    Accepted
    Localpref: 100
    Router ID: 101.0.0.7

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {indirect(1048580)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a5f4 (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65004 I
    Communities:
Path 2008:4498:104:: from 10.0.0.6 Vector len 4. Val: 0
  *BGP
    Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9560a90
    Next-hop reference count: 3
    Source: 10.0.0.6
    Next hop type: Router, Next hop index: 1048578
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
    Label-switched-path LSP-R1-to-R6-1
    Label operation: Push 300208
    Label TTL action: prop-ttl
    Load balance label: Label 300208: None;
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Label-switched-path LSP-R1-to-R6-2
    Label operation: Push 300080
    Label TTL action: prop-ttl
    Load balance label: Label 300080: None;
    Session Id: 0x380005
    Protocol next hop: 2008:4498:610::10
    Indirect next hop: 0x976da3c 1048580 INH Session ID: 0x38000c
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 14:47 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.6+64719
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65004 I
    Accepted
    Localpref: 100
    Router ID: 10.0.0.6
    Indirect next hops: 1
      Protocol next hop: 2008:4498:610::10 Metric: 3
      Indirect next hop: 0x976da3c 1048580 INH Session ID: 0x38000c
      Indirect path forwarding next hops: 2

```

```

                Next hop type: Router
                Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
                Session Id: 0x380004
                Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
                Session Id: 0x380005
2008:4498:610::/120 Originating RIB: inet6.0
                Metric: 3                      Node path count: 1
                Indirect nexthops: 1
                Protocol Nexthop: ::ffff:10.0.0.6 Metric: 3
                Indirect nexthop: 0x976d918 1048579 INH Session ID:
0x38000b                Indirect path forwarding nexthops: 2
                Nexthop: 10.1.2.2 via ge-0/0/1.12

2008:4498:610::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:610::/120 -> {indirect(1048579)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a7d0 (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] I
    Communities:
Path 2008:4498:610:: from 10.0.0.6 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x95627c4
    Next-hop reference count: 3
    Source: 10.0.0.6
    Next hop type: Router, Next hop index: 1048578
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
    Label-switched-path LSP-R1-to-R6-1
    Label operation: Push 300208
    Label TTL action: prop-ttl
    Load balance label: Label 300208: None;
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Label-switched-path LSP-R1-to-R6-2
    Label operation: Push 300080
    Label TTL action: prop-ttl
    Load balance label: Label 300080: None;
    Session Id: 0x380005
    Protocol next hop: ::ffff:10.0.0.6
    Indirect next hop: 0x976d918 1048579 INH Session ID: 0x38000b
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 14:47 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.6+64719
    Announcement bits (4): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1 3-
Resolve_IGP_FRR task
    AS path: I
    Accepted
    Localpref: 100
    Router ID: 10.0.0.6
    Indirect next hops: 1
    Protocol next hop: ::ffff:10.0.0.6 Metric: 3
    Indirect next hop: 0x976d918 1048579 INH Session ID: 0x38000b
    Indirect path forwarding next hops: 2
    Next hop type: Router
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Session Id: 0x380005
    ::ffff:10.0.0.6/128 Originating RIB: inet6.3
    Metric: 3                      Node path count: 1
    Forwarding nexthops: 2
    Nexthop: 10.1.2.2 via ge-0/0/1.12

```

On R6 router

```

lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
  Prefix                Nexthop          MED      Lclpref    AS path
* 2008:4498:104::/64      2008:4498:610::10  100      100        65004 I
* 2008:4498:610::/120    Self              100      100        I

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6 extensive

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Nexthop: 2008:4498:610::10
    Localpref: 100
    AS path: [100] 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Nexthop: Self
    Localpref: 100
    AS path: [100] I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.1 logical-system r6 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

mpls.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  Accepted
  Nexthop: ::ffff:10.0.0.1
  Localpref: 100
  AS path: I

* 2008:4498:101::/64 (1 entry, 1 announced)
  Accepted
  Nexthop: 2008:4498:17::7
  Localpref: 100
  AS path: 65001 I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system r6

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:17::/120 * [BGP/170] 00:24:35, localpref 100, from 10.0.0.1
                  AS path: I, validation-state: unverified
                  > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
                  > to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2
2008:4498:101::/64 * [BGP/170] 00:24:35, localpref 100, from 10.0.0.1
                  AS path: 65001 I, validation-state: unverified
                  > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
                  > to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2
2008:4498:104::/64 * [BGP/170] 08:21:27, localpref 100
                  AS path: 65004 I, validation-state: unverified
                  > to 2008:4498:610::10 via ge-0/0/3.610

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system r6 extensive | no-more

inet6.0: 9 destinations, 11 routes (9 active, 0 holddown, 0 hidden)

```

```

2008:4498:17::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:17::/120 -> {indirect(1048576)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957aae0 (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] I
    Communities:
Path 2008:4498:17:: from 10.0.0.1 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9560a90
    Next-hop reference count: 3
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048575
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
    Label-switched-path LSP-R6-to-R1-1
    Label operation: Push 300176
    Label TTL action: prop-ttl
    Load balance label: Label 300176: None;
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Label-switched-path LSP-R6-to-R1-2
    Label operation: Push 300080
    Label TTL action: prop-ttl
    Load balance label: Label 300080: None;
    Session Id: 0x300004
    Protocol next hop: ::ffff:10.0.0.1
    Indirect next hop: 0x9759918 1048576 INH Session ID: 0x30000e
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 24:38 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.1+179
    Announcement bits (4): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1 3-
Resolve_IGP_FRR task
    AS path: I
    Accepted
    Localpref: 100
    Router ID: 10.0.0.1
    Indirect next hops: 1
      Protocol next hop: ::ffff:10.0.0.1 Metric: 3
      Indirect next hop: 0x9759918 1048576 INH Session ID: 0x30000e
      Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%
        Session Id: 0x300005
        Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
        Session Id: 0x300004
      ::ffff:10.0.0.1/128 Originating RIB: inet6.3
      Metric: 3 Node path count: 1
      Forwarding nexthops: 2
      Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:101::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:101::/64 -> {indirect(1048577)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957990c (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:101:: from 10.0.0.1 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x956247c
    Next-hop reference count: 3
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048575

```

```

Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
Label-switched-path LSP-R6-to-R1-1
Label operation: Push 300176
Label TTL action: prop-ttl
Load balance label: Label 300176: None;
Session Id: 0x300005
Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
Label-switched-path LSP-R6-to-R1-2
Label operation: Push 300080
Label TTL action: prop-ttl
Load balance label: Label 300080: None;
Session Id: 0x300004
Protocol next hop: 2008:4498:17::7
Indirect next hop: 0x9759a3c 1048577 INH Session ID: 0x30000f
State: <Active Int Ext>
Local AS: 100 Peer AS: 100
Age: 24:38 Metric2: 3
Validation State: unverified
Task: BGP_100.10.0.0.1+179
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: 65001 I
Accepted
Localpref: 100
Router ID: 10.0.0.1
Indirect next hops: 1
    Protocol next hop: 2008:4498:17::7 Metric: 3
    Indirect next hop: 0x9759a3c 1048577 INH Session ID: 0x30000f
    Indirect path forwarding next hops: 2
        Next hop type: Router
        Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%
        Session Id: 0x300005
        Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
        Session Id: 0x300004
    2008:4498:17::/120 Originating RIB: inet6.0
        Metric: 3 Node path count: 1
        Indirect nexthops: 1
            Protocol Nexthop: ::ffff:10.0.0.1 Metric: 3
            Indirect nexthop: 0x9759918 1048576 INH Session ID:
Ox30000e                Indirect path forwarding nexthops: 2
                        Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {2008:4498:610::10}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957aa8c (adv_entry)
    Advertised metrics:
        Nexthop: 2008:4498:610::10
        Localpref: 100
        AS path: [100] 65004 I
        Communities:
Path 2008:4498:104:: from 2008:4498:610::10 Vector len 4. Val: 1
    *BGP Preference: 170/-101
        Next hop type: Router, Next hop index: 1267
        Address: 0x9560550
        Next-hop reference count: 3
        Source: 2008:4498:610::10
        Next hop: 2008:4498:610::10 via ge-0/0/3.610, selected
        Session Id: 0x300003
        State: <Active Ext>
        Local AS: 100 Peer AS: 65004
        Age: 8:21:30
        Validation State: unverified
        Task: BGP_65004.2008:4498:610::10+57753
        Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
        AS path: 65004 I
        Accepted
        Localpref: 100
        Router ID: 104.0.0.10

```

Ping test

```
lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
```

```

PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

```

TC6: iBGP family inet6 unicast + enable family inet6 on interfaces on R2, R3 facing R1 and R4, R5 facing R6

Configuration

Continue with the configuration on TC5 and enable family inet6 on interfaces on R2, R3 facing R1 and R4, R5 facing R6

```
[edit logical-systems r2 interfaces ge-0/0/2 unit 12]
+   family inet6;
[edit logical-systems r3 interfaces ge-0/0/2 unit 13]
+   family inet6;
[edit logical-systems r4 interfaces ge-0/0/1 unit 46]
+   family inet6;
[edit logical-systems r5 interfaces ge-0/0/1 unit 56]
+   family inet6;
```

Verification

The same on TC5 so we'll bypass this step for briefly

Ping test

```
lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=60 time=10.522 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.522/10.522/10.522/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=60 time=6.931 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 6.931/6.931/6.931/0.000 ms

[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=60 time=7.188 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.188/7.188/7.188/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=60 time=7.494 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.494/7.494/7.494/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6
```

```
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=62 time=109.742 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 109.742/109.742/109.742/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=62 time=6.255 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 6.255/6.255/6.255/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=61 time=9.286 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.286/9.286/9.286/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=8.863 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.863/8.863/8.863/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=61 time=9.358 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.358/9.358/9.358/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=8.891 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.891/8.891/8.891/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=61 time=27.073 ms
```

```

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 27.073/27.073/27.073/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=7.119 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.119/7.119/7.119/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=61 time=6.002 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 6.002/6.002/6.002/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=11.876 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 11.876/11.876/11.876/0.000 ms

```

TC7: iBGP family inet6 unicast + enable inet6 on R2,R3 facing R1 and R4,R5 facing R6 + RSVP explicit -null

Configuration

Continue with the configuration on TC6 and configure MPLS LSP explicit-null on R1 and R6.

```

lab@vMX3# show | compare
[edit logical-systems r1 protocols mpls]
+   explicit-null;
[edit logical-systems r6 protocols mpls]
+   explicit-null;

```

Verification

The same on TC6 so we'll bypass this step for briefly

Ping test

```

lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

```

```
[edit]
lab@VMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@VMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@VMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
lab@VMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@VMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10
```

```

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@VMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@VMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

[edit]
lab@VMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 0 packets received, 100% packet loss

```

TC8: iBGP family inet6 labeled-unicast + C1, C4 will advertise directed IPv6 range + enable advertise-inactive knob on PE routers (R1, R6)

Configuration

Continue with the configuration on TC2 but we delete configuration concern advertising directed IPv6 range on R1, R6 routers and let C1 and C4 do it, but on R1, R6 we have to enable advertise-inactive as below:

```

lab@VMX3# show | compare rollback 3
[edit logical-systems r1 protocols bgp group ibgp]
+   advertise-inactive;
-   export bgp-exp;
[edit logical-systems r1]
-   policy-options {
-     policy-statement bgp-exp {
-       term 1 {

```

```

-         from {
-             protocol direct;
-             rib inet6.0;
-             route-filter 2008:4498::/32 longer;
-         }
-         then accept;
-     }
- }
[edit logical-systems r6 protocols bgp group ibgp]
+   advertise-inactive;
-   export bgp-exp;
[edit logical-systems r6 policy-options]
-   policy-statement bgp-exp {
-       term 1 {
-           from {
-               protocol direct;
-               rib inet6.0;
-               route-filter 2008:4498::/32 longer;
-           }
-           then accept;
-       }
-   }
[edit policy-options policy-statement c1-bgp-exp]
  term 1 { ... }
+   term 2 {
+       from {
+           protocol direct;
+           rib c1.inet6.0;
+           route-filter 2008:4498::/32 longer;
+       }
+       then accept;
+   }
[edit policy-options policy-statement c4-bgp-exp]
  term 1 { ... }
+   term 2 {
+       from {
+           protocol direct;
+           rib c4.inet6.0;
+           route-filter 2008:4498::/32 longer;
+       }
+       then accept;
+   }

```

Verification

On R1 router

```

lab@vMX3# run show route table inet6.0 logical-system r1 protocol bgp

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:17::/120 [BGP/170] 00:23:31, localpref 100
                  AS path: 65001 I, validation-state: unverified
                  > to 2008:4498:17::7 via ge-0/0/3.17
2008:4498:101::/64 *[BGP/170] 13:08:30, localpref 100
                  AS path: 65001 I, validation-state: unverified
                  > to 2008:4498:17::7 via ge-0/0/3.17
2008:4498:104::/64 *[BGP/170] 00:36:11, localpref 100, from 10.0.0.6
                  AS path: 65004 I, validation-state: unverified
                  > to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-
1
                  to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-
2
2008:4498:610::/120*[BGP/170] 00:22:25, localpref 100, from 10.0.0.6
                  AS path: 65004 I, validation-state: unverified

```

```

> to 10.1.2.2 via ge-0/0/1.12, label-switched-path LSP-R1-to-R6-
1
to 10.1.3.3 via ge-0/0/1.13, label-switched-path LSP-R1-to-R6-
2

[edit]
lab@vMX3# run show route table inet6.0 logical-system r1 2008:4498:17::/120

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:17::/120 *[Direct/0] 23:39:32
> via ge-0/0/3.17
[BGP/170] 00:23:40, localpref 100
AS path: 65001 I, validation-state: unverified
> to 2008:4498:17::7 via ge-0/0/3.17
2008:4498:17::1/128*[Local/0] 23:39:46
Local via ge-0/0/3.17

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
Prefix NextHop MED Lclpref AS path
2008:4498:17::/120 Self 100 65001 I
* 2008:4498:101::/64 Self 100 65001 I

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.6 logical-system r1
extensive

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
2008:4498:17::/120 (2 entries, 2 announced)
BGP group ibgp type Internal
Route Label: 300080
NextHop: Self
Flags: NextHop Change
Localpref: 100
AS path: [100] 65001 I

* 2008:4498:101::/64 (1 entry, 1 announced)
BGP group ibgp type Internal
Route Label: 300080
NextHop: Self
Flags: NextHop Change
Localpref: 100
AS path: [100] 65001 I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.6 logical-system r1 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

mpls.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
Accepted
Route Label: 299920
NextHop: ::ffff:10.0.0.6
Localpref: 100
AS path: 65004 I

* 2008:4498:610::/120 (1 entry, 1 announced)
Accepted
Route Label: 299920
NextHop: ::ffff:10.0.0.6

```

```

Localpref: 100
AS path: 65004 I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 logical-system rl extensive

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
2008:4498:17::/120 (2 entries, 2 announced)
TSI:
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a8e8 (adv_entry)
  Advertised metrics:
    Flags: Nexthop Change
    Nexthop: Self
    Localpref: 100
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:17:: from 2008:4498:17::7 Vector len 4. Val: 1
  *Direct Preference: 0
    Next hop type: Interface
    Address: 0x9562818
    Next-hop reference count: 2
    Next hop: via ge-0/0/3.17, selected
    State: <Active Int>
    Local AS: 100
    Age: 23:41:04
    Validation State: unverified
    Task: IF
    Announcement bits (1): 2-Resolve tree 1
    AS path: I
  BGP Preference: 170/-101
    Next hop type: Router, Next hop index: 1266
    Address: 0x956010c
    Next-hop reference count: 5
    Source: 2008:4498:17::7
    Next hop: 2008:4498:17::7 via ge-0/0/3.17, selected
    Session Id: 0x380003
    State: <Ext>
    Inactive reason: Route Preference
    Local AS: 100 Peer AS: 65001
    Age: 25:12
    Validation State: unverified
    Task: BGP_65001.2008:4498:17::7+63777
    Announcement bits (1): 1-BGP_RT_Background
    AS path: 65001 I
    Accepted
    Localpref: 100
    Router ID: 101.0.0.7

2008:4498:17::1/128 (1 entry, 1 announced)
  *Local Preference: 0
    Next hop type: Local
    Address: 0x92948a4
    Next-hop reference count: 10
    Next hop:
    Interface: ge-0/0/3.17
    State: <Active NoReadvrt Int>
    Local AS: 100
    Age: 23:41:18
    Validation State: unverified
    Task: IF
    Announcement bits (1): 2-Resolve tree 1
    AS path: I

2008:4498:101::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:101::/64 -> {2008:4498:17::7}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957a728 (adv entry)

```

```

Advertised metrics:
  Flags: Nexthop Change
  Nexthop: Self
  Localpref: 100
  AS path: [100] 65001 I
  Communities:
Path 2008:4498:101:: from 2008:4498:17::7 Vector len 4. Val: 1
  *BGP Preference: 170/-101
    Next hop type: Router, Next hop index: 1266
    Address: 0x956010c
    Next-hop reference count: 5
    Source: 2008:4498:17::7
    Next hop: 2008:4498:17::7 via ge-0/0/3.17, selected
    Session Id: 0x380003
    State: <Active Ext>
    Local AS: 100 Peer AS: 65001
    Age: 13:10:11
    Validation State: unverified
    Task: BGP_65001.2008:4498:17::7+63777
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65001 I
    Accepted
    Localpref: 100
    Router ID: 101.0.0.7

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {indirect(1048576)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a38c (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65004 I
    Communities:
Path 2008:4498:104:: from 10.0.0.6 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x9561270
    Next-hop reference count: 6
    Source: 10.0.0.6
    Next hop type: Router, Next hop index: 1048575
    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
    Label-switched-path LSP-R1-to-R6-1
    Label operation: Push 299920, Push 300272(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 299920: None; Label 300272: None;
    Session Id: 0x380004
    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
    Label-switched-path LSP-R1-to-R6-2
    Label operation: Push 299920, Push 300144(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 299920: None; Label 300144: None;
    Session Id: 0x380005
    Protocol next hop: ::ffff:10.0.0.6
    Label operation: Push 299920
    Label TTL action: prop-ttl
    Load balance label: Label 299920: None;
    Indirect next hop: 0x976d918 1048576 INH Session ID: 0x38000d
    State: <Active Int Ext>
    Local AS: 100 Peer AS: 100
    Age: 37:52 Metric2: 3
    Validation State: unverified
    Task: BGP_100.10.0.0.6+179
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65004 I
    Accepted
    Route Label: 299920
    Localpref: 100
    Router ID: 10.0.0.6
    Indirect next hops: 1

```

```

                                Protocol next hop: ::ffff:10.0.0.6 Metric: 3
                                Label operation: Push 299920
                                Label TTL action: prop-ttl
                                Load balance label: Label 299920: None;
                                Indirect next hop: 0x976d918 1048576 INH Session ID:
0x38000d
                                Indirect path forwarding next hops: 2
                                    Next hop type: Router
                                    Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
                                    Session Id: 0x380004
                                    Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
                                    Session Id: 0x380005
                                ::ffff:10.0.0.6/128 Originating RIB: inet6.3
                                Metric: 3                               Node path count: 1
                                Forwarding nexthops: 2
                                    Nexthop: 10.1.2.2 via ge-0/0/1.12

2008:4498:610::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:610::/120 -> {indirect(1048576)}
Page 0 idx 0, (group ebgp-cl-v6 type External) Type 1 val 0x957a70c (adv_entry)
    Advertised metrics:
        Nexthop: Self
        AS path: [100] 65004 I
        Communities:
Path 2008:4498:610:: from 10.0.0.6 Vector len 4. Val: 0
    *BGP      Preference: 170/-101
              Next hop type: Indirect
              Address: 0x9561270
              Next-hop reference count: 6
              Source: 10.0.0.6
              Next hop type: Router, Next hop index: 1048575
              Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1, selected
              Label-switched-path LSP-R1-to-R6-1
              Label operation: Push 299920, Push 300272(top)
              Label TTL action: prop-ttl, prop-ttl(top)
              Load balance label: Label 299920: None; Label 300272: None;
              Session Id: 0x380004
              Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
              Label-switched-path LSP-R1-to-R6-2
              Label operation: Push 299920, Push 300144(top)
              Label TTL action: prop-ttl, prop-ttl(top)
              Load balance label: Label 299920: None; Label 300144: None;
              Session Id: 0x380005
              Protocol next hop: ::ffff:10.0.0.6
              Label operation: Push 299920
              Label TTL action: prop-ttl
              Load balance label: Label 299920: None;
              Indirect next hop: 0x976d918 1048576 INH Session ID: 0x38000d
              State: <Active Int Ext>
              Local AS: 100 Peer AS: 100
              Age: 24:06      Metric2: 3
              Validation State: unverified
              Task: BGP_100.10.0.0.6+179
              Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
              AS path: 65004 I
              Accepted
              Route Label: 299920
              Localpref: 100
              Router ID: 10.0.0.6
              Indirect next hops: 1
                  Protocol next hop: ::ffff:10.0.0.6 Metric: 3
                  Label operation: Push 299920
                  Label TTL action: prop-ttl
                  Load balance label: Label 299920: None;
                  Indirect next hop: 0x976d918 1048576 INH Session ID:
0x38000d
                  Indirect path forwarding next hops: 2
                      Next hop type: Router
```

```

Next hop: 10.1.2.2 via ge-0/0/1.12 weight 0x1
Session Id: 0x380004
Next hop: 10.1.3.3 via ge-0/0/1.13 weight 0x1
Session Id: 0x380005
::ffff:10.0.0.6/128 Originating RIB: inet6.3
Metric: 3 Node path count: 1
Forwarding nexthops: 2
Nexthop: 10.1.2.2 via ge-0/0/1.12

```

On R6 router

```

lab@vMX3# run show route table inet6.0 logical-system r6 protocol bgp

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:17::/120 *[BGP/170] 00:27:52, localpref 100, from 10.0.0.1
    AS path: 65001 I, validation-state: unverified
    > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
    to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2
2008:4498:101::/64 *[BGP/170] 00:41:38, localpref 100, from 10.0.0.1
    AS path: 65001 I, validation-state: unverified
    > to 10.4.6.4 via ge-0/0/2.46, label-switched-path LSP-R6-to-R1-1
    to 10.5.6.5 via ge-0/0/2.56, label-switched-path LSP-R6-to-R1-2
2008:4498:104::/64 *[BGP/170] 13:13:54, localpref 100
    AS path: 65004 I, validation-state: unverified
    > to 2008:4498:610::10 via ge-0/0/3.610
2008:4498:610::/120 [BGP/170] 00:28:58, localpref 100
    AS path: 65004 I, validation-state: unverified
    > to 2008:4498:610::10 via ge-0/0/3.610

[edit]
lab@vMX3# run show route table inet6.0 logical-system r6 2008:4498:610::/120

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:4498:610::/120*[Direct/0] 23:44:58
    > via ge-0/0/3.610
    [BGP/170] 00:29:07, localpref 100
    AS path: 65004 I, validation-state: unverified
    > to 2008:4498:610::10 via ge-0/0/3.610
2008:4498:610::/128
    *[Local/0] 23:45:13
    Local via ge-0/0/3.610

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
Prefix Nexthop MED Lclpref AS path
* 2008:4498:104::/64 Self 100 65004 I
  2008:4498:610::/120 Self 100 65004 I

[edit]
lab@vMX3# run show route advertising-protocol bgp 10.0.0.1 logical-system r6 extensive

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:104::/64 (1 entry, 1 announced)
  BGP group ibgp type Internal
    Route Label: 299920
    Nexthop: Self
    Flags: Nexthop Change
    Localpref: 100
    AS path: [100] 65004 I

  2008:4498:610::/120 (2 entries, 2 announced)
    BGP group ibgp type Internal

```

```

Route Label: 299920
Nexthop: Self
Flags: Nexthop Change
Localpref: 100
AS path: [100] 65004 I

[edit]
lab@vMX3# run show route receive-protocol bgp 10.0.0.1 logical-system r6 extensive

inet.0: 17 destinations, 17 routes (17 active, 0 holddown, 0 hidden)

inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

mpls.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
* 2008:4498:17::/120 (1 entry, 1 announced)
  Accepted
  Route Label: 300080
  Nexthop: ::ffff:10.0.0.1
  Localpref: 100
  AS path: 65001 I

* 2008:4498:101::/64 (1 entry, 1 announced)
  Accepted
  Route Label: 300080
  Nexthop: ::ffff:10.0.0.1
  Localpref: 100
  AS path: 65001 I

inet6.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)

[edit]
lab@vMX3# run show route table inet6.0 protocol bgp logical-system r6 extensive

inet6.0: 9 destinations, 12 routes (9 active, 0 holddown, 0 hidden)
2008:4498:17::/120 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:17::/120 -> {indirect(1048582)}
Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957a3fc (adv_entry)
  Advertised metrics:
    Nexthop: Self
    AS path: [100] 65001 I
    Communities:
Path 2008:4498:17:: from 10.0.0.1 Vector len 4. Val: 0
  *BGP Preference: 170/-101
    Next hop type: Indirect
    Address: 0x940de94
    Next-hop reference count: 6
    Source: 10.0.0.1
    Next hop type: Router, Next hop index: 1048581
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
    Label-switched-path LSP-R6-to-R1-1
    Label operation: Push 300080, Push 300240(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 300080: None; Label 300240: None;
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Label-switched-path LSP-R6-to-R1-2
    Label operation: Push 300080, Push 300144(top)
    Label TTL action: prop-ttl, prop-ttl(top)
    Load balance label: Label 300080: None; Label 300144: None;
    Session Id: 0x300004
    Protocol next hop: ::ffff:10.0.0.1
    Label operation: Push 300080
    Label TTL action: prop-ttl
    Load balance label: Label 300080: None;
    Indirect next hop: 0x9759a3c 1048582 INH Session ID: 0x300010
    State: <Active Int Ext>

```

```
Local AS: 100 Peer AS: 100
Age: 28:42 Metric2: 3
Validation State: unverified
Task: BGP_100.10.0.0.1+56184
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: 65001 I
Accepted
Route Label: 300080
Localpref: 100
Router ID: 10.0.0.1
Indirect next hops: 1
  Protocol next hop: ::ffff:10.0.0.1 Metric: 3
  Label operation: Push 300080
  Label TTL action: prop-ttl
  Load balance label: Label 300080: None;
  Indirect next hop: 0x9759a3c 1048582 INH Session ID: 0x300010
  Indirect path forwarding next hops: 2
    Next hop type: Router
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Session Id: 0x300004
  ::ffff:10.0.0.1/128 Originating RIB: inet6.3
  Metric: 3 Node path count: 1
  Forwarding nexthops: 2
  Nexthop: 10.4.6.4 via ge-0/0/2.46
```

2008:4498:101::/64 (1 entry, 1 announced)

TSI:

KRT in-kernel 2008:4498:101::/64 -> {indirect(1048582)}

Page 0 idx 0, (group ebgp-c4-v6 type External) Type 1 val 0x957ab6c (adv_entry)

Advertised metrics:

Nexthop: Self

AS path: [100] 65001 I

Communities:

Path 2008:4498:101:: from 10.0.0.1 Vector len 4. Val: 0

```
*BGP Preference: 170/-101
Next hop type: Indirect
Address: 0x940de94
Next-hop reference count: 6
Source: 10.0.0.1
Next hop type: Router, Next hop index: 1048581
Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%, selected
Label-switched-path LSP-R6-to-R1-1
Label operation: Push 300080, Push 300240(top)
Label TTL action: prop-ttl, prop-ttl(top)
Load balance label: Label 300080: None; Label 300240: None;
Session Id: 0x300005
Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
Label-switched-path LSP-R6-to-R1-2
Label operation: Push 300080, Push 300144(top)
Label TTL action: prop-ttl, prop-ttl(top)
Load balance label: Label 300080: None; Label 300144: None;
Session Id: 0x300004
Protocol next hop: ::ffff:10.0.0.1
Label operation: Push 300080
Label TTL action: prop-ttl
Load balance label: Label 300080: None;
Indirect next hop: 0x9759a3c 1048582 INH Session ID: 0x300010
State: <Active Int Ext>
Local AS: 100 Peer AS: 100
Age: 42:28 Metric2: 3
Validation State: unverified
Task: BGP_100.10.0.0.1+56184
Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
AS path: 65001 I
Accepted
Route Label: 300080
Localpref: 100
```

```

Router ID: 10.0.0.1
Indirect next hops: 1
  Protocol next hop: ::ffff:10.0.0.1 Metric: 3
  Label operation: Push 300080
  Label TTL action: prop-ttl
  Load balance label: Label 300080: None;
  Indirect next hop: 0x9759a3c 1048582 INH Session ID: 0x300010
  Indirect path forwarding next hops: 2
    Next hop type: Router
    Next hop: 10.4.6.4 via ge-0/0/2.46 weight 0x1 balance 33%
    Session Id: 0x300005
    Next hop: 10.5.6.5 via ge-0/0/2.56 weight 0x1 balance 67%
    Session Id: 0x300004
  ::ffff:10.0.0.1/128 Originating RIB: inet6.3
  Metric: 3 Node path count: 1
  Forwarding nexthops: 2
    Nexthop: 10.4.6.4 via ge-0/0/2.46

2008:4498:104::/64 (1 entry, 1 announced)
TSI:
KRT in-kernel 2008:4498:104::/64 -> {2008:4498:610::10}
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957990c (adv_entry)
  Advertised metrics:
    Flags: Nexthop Change
    Nexthop: Self
    Localpref: 100
    AS path: [100] 65004 I
    Communities:
Path 2008:4498:104:: from 2008:4498:610::10 Vector len 4. Val: 1
  *BGP Preference: 170/-101
    Next hop type: Router, Next hop index: 1267
    Address: 0x9560550
    Next-hop reference count: 5
    Source: 2008:4498:610::10
    Next hop: 2008:4498:610::10 via ge-0/0/3.610, selected
    Session Id: 0x300003
    State: <Active Ext>
    Local AS: 100 Peer AS: 65004
    Age: 13:14:44
    Validation State: unverified
    Task: BGP_65004.2008:4498:610::10+57753
    Announcement bits (3): 0-KRT 1-BGP_RT_Background 2-Resolve tree 1
    AS path: 65004 I
    Accepted
    Localpref: 100
    Router ID: 104.0.0.10

2008:4498:610::/120 (2 entries, 2 announced)
TSI:
Page 0 idx 1, (group ibgp type Internal) Type 1 val 0x957aa8c (adv_entry)
  Advertised metrics:
    Flags: Nexthop Change
    Nexthop: Self
    Localpref: 100
    AS path: [100] 65004 I
    Communities:
Path 2008:4498:610:: from 2008:4498:610::10 Vector len 4. Val: 1
  BGP Preference: 170/-101
    Next hop type: Router, Next hop index: 1267
    Address: 0x9560550
    Next-hop reference count: 5
    Source: 2008:4498:610::10
    Next hop: 2008:4498:610::10 via ge-0/0/3.610, selected
    Session Id: 0x300003
    State: <Ext>
    Inactive reason: Route Preference
    Local AS: 100 Peer AS: 65004
    Age: 29:48
    Validation State: unverified

```

```
Task: BGP_65004.2008:4498:610::10+57753
Announcement bits (1): 1-BGP_RT_Background
AS path: 65004 I
Accepted
Localpref: 100
Router ID: 104.0.0.10
```

```
[edit]
lab@vMX3#
```

Ping test

```
lab@vMX3# ##Ping between C1-C4 lo0 (1)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 routing-instance c1 source 2008:4498:101::7
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=60 time=7.279 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.279/7.279/7.279/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:101::7 routing-instance c4 source 2008:4498:104::10
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=60 time=7.692 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.692/7.692/7.692/0.000 ms

[edit]
lab@vMX3# ##Ping between C1-C4 p2p-wan (2)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=60 time=7.773 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.773/7.773/7.773/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::7 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=60 time=6.717 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 6.717/6.717/6.717/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-R6 p2p-wan (3)-(4)

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:17::1 logical-system r1 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=60 time=10.337 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.337/10.337/10.337/0.000 ms
```

```
[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=60 time=8.787 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.787/8.787/8.787/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 p2p-wan (3)-(5)

[edit]
lab@vMX3# run ping 2008:4498:610::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:610::10
16 bytes from 2008:4498:610::10, icmp_seq=0 hlim=59 time=8.830 ms

--- 2008:4498:610::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.830/8.830/8.830/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 routing-instance c4 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=6.186 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 6.186/6.186/6.186/0.000 ms

[edit]
lab@vMX3# ##Ping between R1-C4 wan-lo0 (3)-(6)

[edit]
lab@vMX3# run ping 2008:4498:104::10 source 2008:4498:17::1 logical-system r1 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::1 --> 2008:4498:104::10
16 bytes from 2008:4498:104::10, icmp_seq=0 hlim=59 time=9.742 ms

--- 2008:4498:104::10 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.742/9.742/9.742/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:17::1 source 2008:4498:104::10 routing-instance c4
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:104::10 --> 2008:4498:17::1
16 bytes from 2008:4498:17::1, icmp_seq=0 hlim=61 time=9.846 ms

--- 2008:4498:17::1 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 9.846/9.846/9.846/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(2)

[edit]
lab@vMX3# run ping 2008:4498:17::7 source 2008:4498:610::6 logical-system r6 count 1
wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:17::7
16 bytes from 2008:4498:17::7, icmp_seq=0 hlim=59 time=10.611 ms

--- 2008:4498:17::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 10.611/10.611/10.611/0.000 ms

[edit]
```

```
lab@vMX3# run ping 2008:4498:610::6 routing-instance c1 count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:17::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=8.028 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 8.028/8.028/8.028/0.000 ms

[edit]
lab@vMX3# ##Ping between R6-C1 (4)-(1)

[edit]
lab@vMX3# run ping 2008:4498:101::7 source 2008:4498:610::6 logical-system r6 count
1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:610::6 --> 2008:4498:101::7
16 bytes from 2008:4498:101::7, icmp_seq=0 hlim=59 time=7.636 ms

--- 2008:4498:101::7 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 7.636/7.636/7.636/0.000 ms

[edit]
lab@vMX3# run ping 2008:4498:610::6 source 2008:4498:101::7 routing-instance c1
count 1 wait 1
PING6(56=40+8+8 bytes) 2008:4498:101::7 --> 2008:4498:610::6
16 bytes from 2008:4498:610::6, icmp_seq=0 hlim=61 time=11.658 ms

--- 2008:4498:610::6 ping6 statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max/std-dev = 11.658/11.658/11.658/0.000 ms
```